Bachelor Thesis

Deterministic Diffusion Models for Image Restoration: From Probabilistic Generation to Cold Diffusion

Villa Giulia
Artificial Intelligence

Academic Year 2024/2025

Bachelor in Artificial Intelligence
Pavia 2025

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Abstract

Over the past decade, deep generative models have profoundly transformed the field of Artificial Intelligence, enabling new paradigms for data synthesis, restoration, and representation. This thesis investigates the theoretical foundations and experimental implementation of Cold Diffusion, a recent deterministic reformulation of diffusion-based generative modeling proposed by Bansal et al. [2].

The work is organized into two main parts. The first develops a theoretical analysis that traces the evolution from Autoencoders and Variational Autoencoders (VAE) [17, 11] to Generative Adversarial Networks (GAN) [6] and classical Diffusion Models [8], highlighting how reconstruction, probabilistic inference, and iterative denoising converge within the deterministic perspective introduced by Cold Diffusion [2].

The second part presents a complete PyTorch implementation of the Cold Diffusion framework applied to image restoration tasks on the CelebA-HQ dataset. Two deterministic degradation operators were considered: a Gaussian blurring process with progressively increasing variance, and a colour-removal (DeColor) operator that gradually suppresses chromatic information across diffusion steps. For each degradation trajectory, a compact U-Net was trained to approximate the inverse mapping, enabling the reconstruction of clean images from their degraded counterparts through an iterative inversion scheme, following the residual correction update introduced in Cold Diffusion [2].

Through both analytical discussion and empirical experimentation, the thesis argues that Cold Diffusion offers a coherent reinterpretation of diffusion-based generative modeling. By replacing stochastic noise with deterministic degradations, the framework unifies the interpretability typical of reconstruction-based models with the stepwise refinement characteristic of diffusion processes, reframing generative modeling as the inversion of structured degradation operators rather than stochastic sampling.

Keywords

Cold Diffusion, Deterministic Diffusion Models, Generative Models, Image Restoration, Gaussian Blur, Colour Degradation

Chapter 1
Introduction

1.1 Background and Motivation

Generative models constitute one of the most conceptually rich and rapidly evolving areas of modern machine learning. Whereas traditional discriminative models learn conditional relationships of the form $p(y \mid x)$, generative models aim to approximate the underlying data distribution $p(x)$ that governs the observed data. Once this distribution is learned, the model can generate new samples that resemble the training data, enabling applications in image synthesis, restoration, speech generation, and data augmentation.

From a broader perspective, the development of deep generative modelling can be interpreted as a progressive refinement of how information is encoded, reconstructed, and sampled. Several methodological milestones have shaped this evolution:

- **Autoencoders (AE)** introduced learned latent representations through the joint optimisation of an encoder–decoder pair [17].

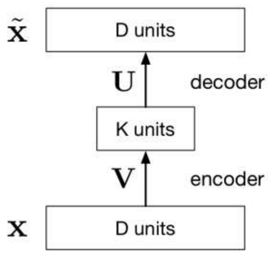

Figure 1.1: Schematic structure of a basic Autoencoder, where the encoder maps the input x to a latent representation through matrix V, and the decoder reconstructs $\tilde{x}$ via matrix U. This illustrates the foundational encoder–decoder paradigm underlying modern generative models.

- **Variational Autoencoders (VAE)** extended this idea by integrating Bayesian inference, enabling sampling from structured latent spaces and providing a principled probabilistic interpretation [11].

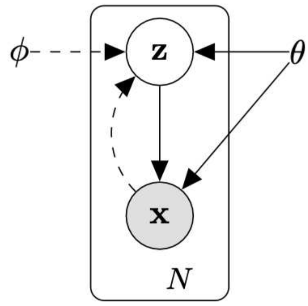

Figure 1.2: Graphical model of a Variational Autoencoder, showing the latent variable z and the generative mapping $p_\theta(x \mid z)$ (adapted from Kingma & Welling, 2013 [11]).

- **Generative Adversarial Networks (GAN)** reframed generative modelling as a two-player min-max game, significantly improving perceptual fidelity through adversarial training [6, 1].

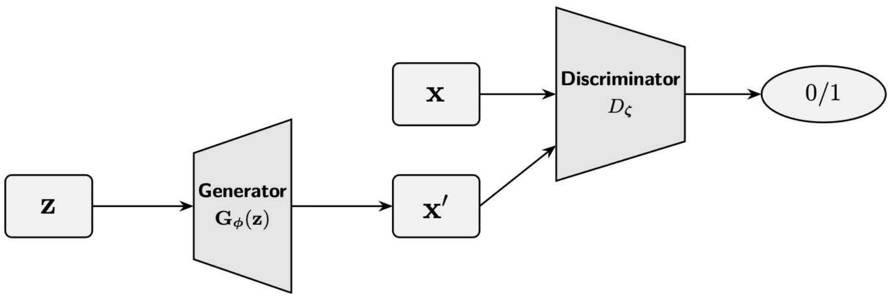

Figure 1.3: Schematic overview of a Generative Adversarial Network (GAN): the generator $G_\phi(z)$ maps a latent vector z to a synthesized sample x', while the discriminator D_ζ distinguishes real samples x from generated ones, producing a binary feedback signal. This adversarial interplay defines the min–max training objective.

- **Diffusion Models** formulated generation as the inversion of a corruption process, typically implemented through Gaussian noise and stochastic sampling [12, 8].

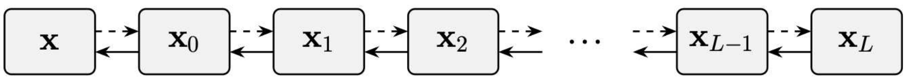

Figure 1.4: Schematic representation of a Diffusion Model (DM): an input sample x is progressively corrupted through a forward trajectory $\{x_t\}_{t=0}^{L}$, while generation corresponds to inverting this process step by step through the learned reverse dynamics.

Cold Diffusion, introduced by Bansal et al. [2], departs from this stochastic formulation. Rather than injecting noise, the forward diffusion process is replaced by a deterministic degradation operator, such as blurring or progressive colour removal, and a neural network is trained to invert this transformation step by step. This perspective shifts the focus from stochastic sampling to the reconstruction of structured degradation trajectories, offering a new viewpoint on what constitutes a generative diffusion process.

From the standpoint of image restoration, this deterministic strategy is particularly appealing. In many real-world scenarios, degradations such as blur, downsampling, or colour desaturation arise from structured physical or algorithmic processes rather than from stochastic Gaussian perturbations. Cold Diffusion directly models these structured transformations as the forward process and learns to invert them iteratively. In doing so, it maintains the gradual refinement characteristic of diffusion models while improving interpretability and alignment with realistic degradation phenomena.

As Richard Feynman famously stated, "What I cannot create, I do not understand." In the context of generative modelling, this idea can be naturally reversed: what we understand, we can create. Deep generative models embody this principle, learning to internalise the structure of data well enough to reproduce it, manipulate it, and ultimately generalise beyond the examples they were trained on.

1.2 Research Objectives and Contributions

This thesis addresses the study of deterministic diffusion models by integrating a rigorous theoretical analysis with a systematic empirical investigation. The work pursues a dual objective. On the theoretical side, it provides a structured overview of the evolution of deep generative modelling, tracing the progression from Autoencoders to Variational Autoencoders, Generative Adversarial Networks, and Diffusion Models, and clarifying how Cold Diffusion arises as a deterministic reinterpretation of the diffusion paradigm. This conceptual synthesis highlights the shared principles underlying likelihood-based and score-based approaches, thereby situating Cold Diffusion within the broader generative modelling landscape.

On the practical side, the thesis develops, implements, and evaluates two Cold Diffusion pipelines for image restoration on the CelebA-HQ dataset. Specifically, it investigates (i) a Gaussian blur operator with increasing variance and (ii) a colour-removal operator that progressively suppresses chromatic components. For both degradation processes, a compact U-Net is trained to approximate the inverse mapping, and reconstruction quality is assessed through iterative inversion following the residual-correction scheme introduced by Bansal et al. This design enables a controlled examination of how deterministic degradation trajectories influence reconstruction dynamics.

The empirical study further contributes a comparative analysis of operator invertibility, contrasting the nearly reversible desaturation process with the partially irreversible blur operator. This comparison elucidates how the structural properties of the forward process shape the behaviour and limitations of deterministic diffusion inversion. Additionally, the implemented pipelines are evaluated against classical restoration baselines - such as Wiener filtering and sharpening - thereby identifying conditions under which deterministic diffusion provides systematic performance advantages.

Overall, the thesis offers (i) a unified conceptual framework for understanding modern generative paradigms, (ii) a complete PyTorch implementation of two deterministic diffusion models, and (iii) an empirical assessment of their reconstruction capabilities and limitations. Through this integration of theory and practice, the work demonstrates how deterministic diffusion models challenge the presumed necessity of stochastic noise in generative processes and underscores their applicability to real-world image restoration tasks.

1.3 Thesis Structure

The remainder of this thesis is organised as outlined below:

Chapter 2 – Theoretical Background: Reviews the evolution of deep generative models, from Autoencoders and Variational Autoencoders [17, 11] to GANs [6, 1, 4] and Diffusion Models [12, 8], providing the conceptual foundations required to understand Cold Diffusion.

Chapter 3 – The Cold Diffusion Framework: Introduces the deterministic formulation proposed by Bansal et al. [2, 3], presenting its mathematical structure and contrasting it with stochastic diffusion models such as DDPM [8] and DDIM.

Chapter 4 – Implementation: Describes the experimental setup developed in this work, including dataset preparation, the definition of deterministic degradation operators, the U-Net architecture, and the training strategy.

Chapter 5 – Results and Discussion: Reports quantitative and qualitative results obtained from the two deterministic diffusion pipelines, analyses reconstruction behaviour for each degradation type, and discusses strengths and limitations of deterministic diffusion.

Chapter 6 – Conclusions and Future Work: Summarises the main contributions of the thesis and outlines possible directions for extending deterministic diffusion models in future research.

Chapter 2
Theoretical Background

Deep generative modeling has grown into one of the central research areas of modern machine learning, providing the conceptual foundations and algorithmic strategies that allow models to synthesize realistic visual data. Over the past decade, the field has matured through several methodological transitions: from early latent-variable formulations such as autoencoders [17], to probabilistic generative models inspired by Bayesian inference such as Variational Autoencoders [11], to adversarial training schemes centered on competition between neural networks in Generative Adversarial Networks [6, 1, 4], and ultimately to diffusion-based approaches grounded in iterative denoising [12, 8]. More recently, the emergence of deterministic diffusion models has challenged the assumption that stochasticity is essential for generative processes, offering a new perspective particularly suited to image restoration tasks, as demonstrated in the Cold Diffusion framework [2, 3].

This chapter develops a unified and discursive overview of these frameworks, emphasizing conceptual continuity and the motivations that progressively led from classical models to the fully deterministic Cold Diffusion paradigm explored in this thesis.

2.1 Introduction to Generative Modeling

A generative model aims to reproduce the underlying data distribution $p_{\text{data}}(x)$ by learning a parametric distribution $p_\theta(x)$. Once trained, the model can draw new samples

$$x' \sim p_\theta(x),$$

that retain the essential characteristics of the dataset. The central training principle for explicit generative models is *maximum likelihood estimation*, which seeks parameters

$$\theta^\star = \arg\max_\theta \mathbb{E}_{x \sim p_{\text{data}}}[\log p_\theta(x)],$$

ensuring that the learned distribution converges, in the Kullback–Leibler sense, toward the true data distribution.

More precisely, maximum likelihood minimises the forward Kullback–Leibler divergence $D_{\text{KL}}(p_{\text{data}} \,\|\, p_\theta)$, ensuring that the learned model assigns high probability mass to the data distribution.

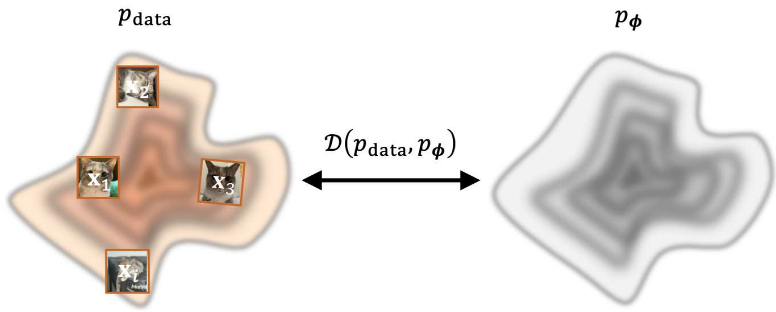

Figure 2.1: Conceptual illustration of deep generative modeling. The model distribution p_θ is trained to approximate the true data distribution p_{data} by minimising a divergence $\mathcal{D}(p_{\text{data}}, p_\theta)$.

Generative models are often categorized according to whether they define a tractable density. **Explicit models**, including Variational Autoencoders (VAEs) [11] and Normalizing Flows, build an explicit and differentiable form of $p_\theta(x)$, allowing principled likelihood-based training. **Implicit models**, such as Generative Adversarial Networks (GANs) [6, 1, 4], sidestep density estimation entirely: they define a sampling mechanism and rely on adversarial feedback to guide the generator toward realism.

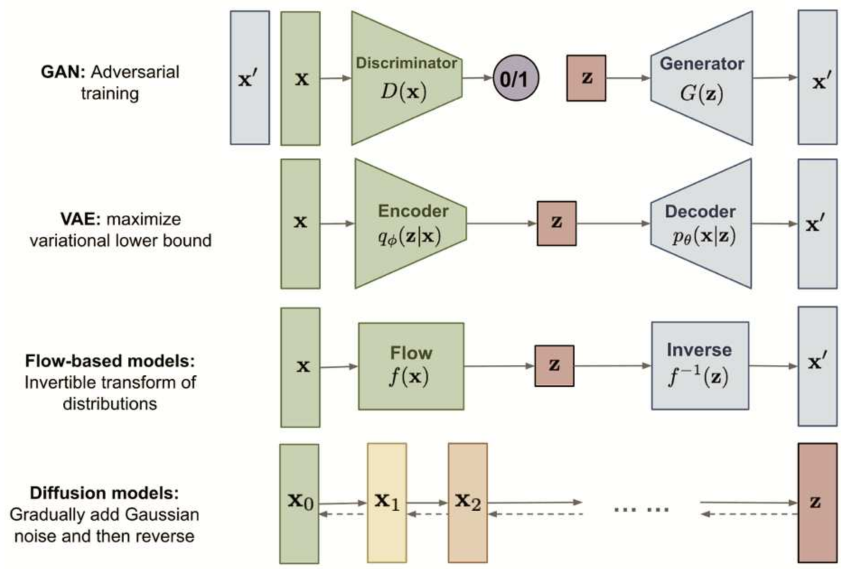

Figure 2.2: Overview of major deep generative model families.

These two paradigms illustrate a persistent tension between interpretability and perceptual quality. Diffusion models have emerged as a compelling compromise: they retain the interpretability of likelihood-based formulations while achieving visual fidelity comparable to adversarial approaches [12, 8, 15] . Their core idea, the iterative inversion of a corruption process, will reappear, in a deterministic form, at the heart of the Cold Diffusion framework [2, 3].

2.2 Autoencoders and Denoising Autoencoders

Autoencoders (AEs) represent one of the earliest attempts to learn lower-dimensional representations of data in an unsupervised fashion [17]. The architecture consists of an encoder $E_\theta(x)$, which projects the input $x \in \mathbb{R}^d$ to a latent vector z, and a decoder $D_\phi(z)$, which reconstructs the original input:

$$\hat{x} = D_\phi(E_\theta(x)).$$

The objective is to minimise the reconstruction error

$$\mathcal{L}_{AE} = \|x - \hat{x}\|^2,$$

encouraging the model to capture the most salient and informative structure of the data. When the latent representation has lower dimensionality than the input, the network is forced to compress the information content of x, and the bottleneck acts as a learned nonlinear feature extractor.

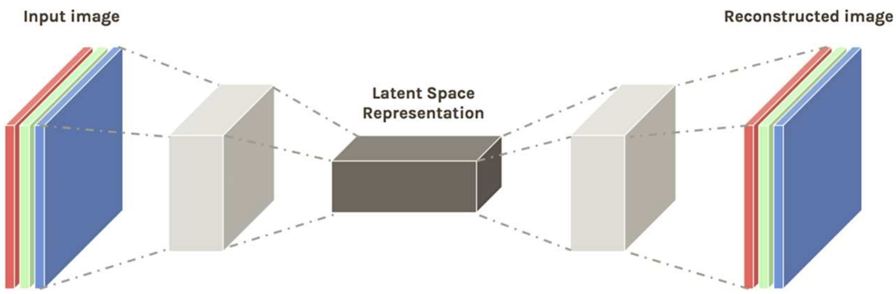

Figure 2.3: Schematic architecture of a standard autoencoder, where the encoder E_θ maps the input to a latent code z and the decoder D_ϕ reconstructs $\hat{x}$.

Despite their effectiveness for dimensionality reduction and representation learning, standard autoencoders lack meaningful generative capabilities: the latent space is typically irregular, and randomly sampling from it rarely yields realistic images. Since the encoder is deterministic, the distribution of latent codes induced by real data is highly structured and supported on a complex manifold; most randomly sampled points fall outside this region and are therefore decoded to implausible outputs. This limitation motivates probabilistic extensions such as the Variational Autoencoder, introduced in Section 2.3.

Denoising Autoencoders (DAEs) introduce a crucial refinement by corrupting the input with Gaussian noise,

$$\tilde{x} = x + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I),$$

and training the network to reconstruct the clean data:

$$\mathcal{L}_{DAE} = \|x - D_\phi(E_\theta(\tilde{x}))\|^2.$$

This encourages the model to learn the structure of the data manifold and to understand, implicitly, how to move noisy inputs back toward high-density regions. This principle directly anticipates the role of noise prediction in modern diffusion models [16, 12, 8].

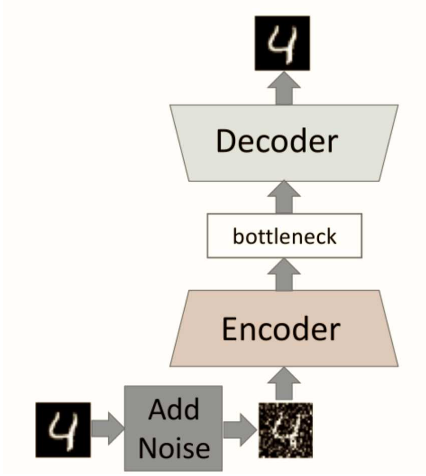

Figure 2.4: Denoising Autoencoder: the input is corrupted with noise and reconstructed through an encoder-decoder architecture E_θ, D_ϕ.

2.3 Variational Autoencoders (VAE)

Variational Autoencoders (VAEs) extend the standard autoencoder architecture by introducing a probabilistic formulation of the latent space [11]. Whereas classical autoencoders map each input to a single deterministic latent vector, VAEs aim to model the underlying variability of the data by associating each input x with a *distribution* over possible latent representations.

More concretely, the encoder no longer outputs a point in latent space, but the parameters of an approximate posterior distribution

$$q_\theta(z \mid x) = \mathcal{N}\big(z;\, \mu_\theta(x),\, \sigma_\theta^2(x)\, I\big),$$

where the neural network produces a mean $\mu_\theta(x)$ and a standard deviation $\sigma_\theta(x)$. Sampling from this distribution allows the model to capture uncertainty and to represent multiple plausible latent codes for the same input.

To complete the generative process, VAEs posit a simple prior over the latent variables,

$$p(z) = \mathcal{N}(0, I),$$

and define a likelihood model $p_\phi(x \mid z)$ - implemented by the decoder - which specifies how a latent sample generates an observation. Together, prior and likelihood define the joint model

$$p_\phi(x, z) = p(z)\, p_\phi(x \mid z),$$

providing a probabilistic interpretation of both encoding and decoding.

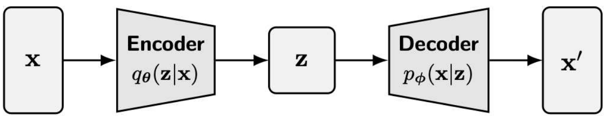

Figure 2.5: Graphical representation of a Variational Autoencoder. The encoder defines $q_\theta(z \mid x)$ and the decoder specifies $p_\phi(x \mid z)$ [11].

In principle, learning would require evaluating the true posterior $p_\phi(z \mid x)$, but this distribution is intractable for most neural decoders. VAEs address this difficulty by maximising the *Evidence Lower Bound* (ELBO),

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\theta(z|x)}[\log p_\phi(x \mid z)] - D_{\text{KL}}\big(q_\theta(z \mid x) \,\|\, p(z)\big)\,,$$

which can be interpreted as balancing two complementary objectives. The first term encourages the decoder to reconstruct the input accurately, while the second term regularises the latent space by penalising deviations of the approximate posterior from the prior. This trade-off induces a smooth, continuous latent manifold from which coherent samples can be generated.

A practical challenge arises from the sampling operation inside the expectation: naively drawing $z \sim q_\theta(z \mid x)$ would break the computational graph. The *reparameterization trick* solves this by expressing sampling as a deterministic transformation of noise:

$$z = \mu_\theta(x) + \sigma_\theta(x) \odot \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I),$$

which enables gradient-based optimisation while preserving stochasticity in the latent space.

Over time, several extensions have enriched the VAE framework. The β-VAE [7] introduces a weighting factor on the KL term, strengthening the pressure toward disentangled latent representations. Conditional VAEs (CVAEs) [13], in turn, incorporate conditioning variables - such as class labels or auxiliary attributes - allowing controlled and targeted generation.

The probabilistic principles underlying VAEs, such as the structured latent space, the role of injected Gaussian noise, and the reconstruction-regularisation trade-off, foreshadow key mechanisms later used in diffusion models. Diffusion-based generative methods reinterpret sampling as the inversion of a corruption process, establishing a conceptual bridge between variational modelling and the noise-prediction viewpoint explored in Section 2.5.

2.4 Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs), introduced by Goodfellow et al. [6], reformulate generative modeling as a two-player min-max game between two neural networks: a generator G and a discriminator D. The generator maps a latent variable $z \sim p(z)$, typically drawn from a simple prior such as a Gaussian distribution, to a synthetic data sample $G(z)$. The discriminator receives either real samples $x \sim p_{\text{data}}$ or generated samples $G(z)$ and outputs the probability that its input is real.

Training proceeds by solving the adversarial objective

$$\min_G \max_D \left[\mathbb{E}_{x \sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z \sim p(z)}[\log(1 - D(G(z)))] \right],$$

in which the discriminator aims to distinguish real from generated samples, while the generator attempts to produce samples that the discriminator classifies as real. At the theoretical optimum, and under idealised assumptions of infinite model capacity, the generator recovers the true data distribution.

Over the years, several architectural and methodological advances have substantially improved sample fidelity and training stability. Notable examples include Wasserstein GANs (WGAN) [1], which introduce a more stable learning objective based on optimal transport; BigGAN [4], which scales adversarial training to large batch sizes and model capacities; and StyleGAN [10], which achieves state-of-the-art photorealism by disentangling latent factors through a style-based architecture.

Despite these successes, GANs present inherent limitations. A major issue is *training instability*: the minimax dynamics often lead to oscillatory behaviour, vanishing gradients, or failure to converge. Moreover, GANs do not define an explicit likelihood, preventing the evaluation of $p(x)$ and limiting their applicability in tasks that require density estimation or probabilistic reasoning.

A particularly well-known failure mode is mode collapse. In this phenomenon, the generator learns to produce samples from only a restricted subset of the modes of the true distribution. As a result, it may repeatedly generate visually plausible yet highly similar outputs, neglecting large regions of the data manifold. This behavior originates from the adversarial training mechanism: if a limited set of samples suffices to fool the discriminator, the generator receives little incentive to model the full diversity of the data. Consequently, GANs often trade sample diversity for perceptual fidelity.

In contrast to GANs, diffusion models avoid adversarial optimisation altogether, replacing the unstable min-max dynamics with a stable, likelihood-inspired denoising objective. These limitations have motivated alternative generative paradigms that aim to combine high-quality synthesis with stability, coverage of the data distribution, and interpretability; likelihood-based models such as Variational Autoencoders, and more recently diffusion models, address several of these shortcomings by providing principled training objectives and more reliable optimisation landscapes. Diffusion models, in particular, achieve GAN-level visual fidelity while offering a clearer probabilistic formulation, making them a compelling successor to adversarial generative approaches.

2.5 Diffusion Models

Diffusion models formulate generative modelling as the inversion of a progressive corruption process applied to clean data [12]. In the forward process, a clean image x_0 is gradually transformed into increasingly noisy versions $\{x_t\}_{t=1}^{T}$ through a Gaussian Markov chain,

$$q(x_t \mid x_{t-1}) = \mathcal{N}\Big(\sqrt{1-\beta_t}\, x_{t-1},\, \beta_t I\Big),$$

where the noise schedule $\{\beta_t\}_{t=1}^{T}$ regulates the rate at which information is destroyed [8]. As $t \to T$, the marginal distribution $q(x_T)$ approaches an isotropic Gaussian independent of the original data.

The generative task consists of reversing this trajectory. However, the true reverse transition

$$q(x_{t-1} \mid x_t)$$

is intractable, making exact inversion impossible. DDPMs therefore introduce a learned approximation of the form

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\Big(\mu_\theta(x_t, t),\, \sigma_t^2 I\Big),$$

where a neural network predicts the mean of the reverse distribution [8]. This modelling choice is justified through a variational lower bound, which provides a tractable objective for approximating the unknown transition.

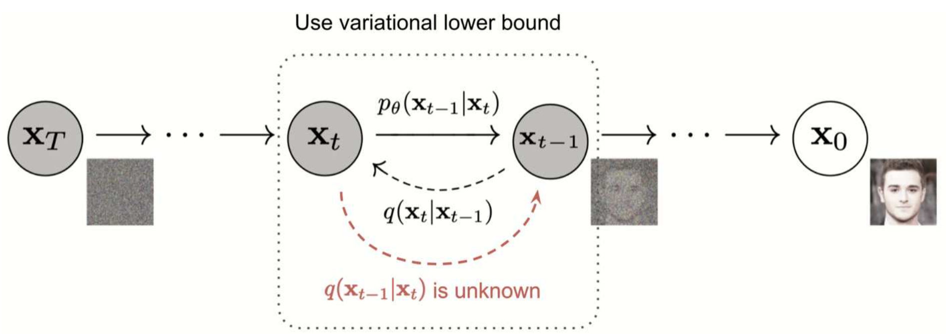

Figure 2.6: Forward and reverse processes in DDPMs. The forward diffusion $q(x_t \mid x_{t-1})$ produces progressively noisier samples, while the true reverse transition $q(x_{t-1} \mid x_t)$ is intractable because its parameters depend on the unknown data distribution. A variational lower bound enables its approximation through the learned model $p_\theta(x_{t-1} \mid x_t)$.

Under the DDPM reparameterization, training simplifies to predicting the noise added during the forward process. The resulting objective,

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0,\,\varepsilon,\,t}\left[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2\right],$$

closely corresponds to score matching [15] and has become the standard formulation used in practice. This approach yields high-quality samples and stable optimization, though sampling typically requires hundreds of reverse-diffusion steps.

Beyond its probabilistic interpretation, DDPM sampling can also be viewed through an energy-based perspective: each reverse-diffusion update defines a vector field guiding samples from a Gaussian prior toward regions of higher data density. This viewpoint provides geometric intuition for why noise-prediction models effectively parameterize the score of the data distribution.

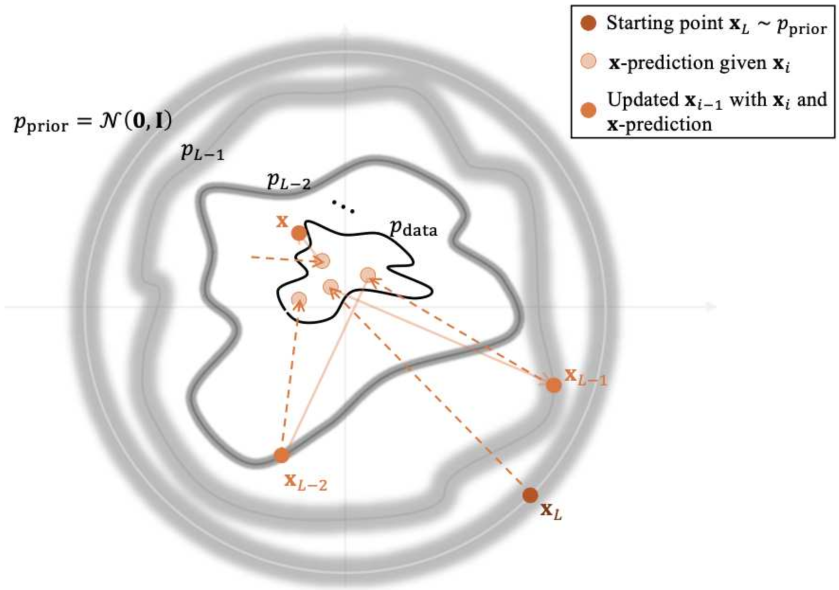

Figure 2.7: Energy-based interpretation of DDPM sampling. Starting from a latent variable $x_L \sim \mathcal{N}(0, I)$, the model iteratively predicts the clean signal x_0 from each noisy state x_i, producing a vector field that transports samples toward regions of high data density.

This geometric perspective naturally motivates the connection with score-based generative modelling, discussed in the next section.

2.6 Score-Based and Energy-Based Perspectives

A complementary and conceptually rich viewpoint on diffusion models arises from Energy-Based Models (EBMs) and the framework of score matching [9]. While a full introduction to stochastic calculus and differential equations is beyond the scope of this thesis, the following section summarises the essential conceptual tools needed to understand how stochastic diffusion models relate to deterministic formulations such as the Probability Flow ODE. The goal is not to provide formal derivations, but to highlight the intuitions that motivate the transition from noisy to fully deterministic diffusion dynamics.

Diffusion models explicitly construct a noisy forward process and learn its probabilistic reversal. Score-based approaches, instead, focus on estimating the score function

$$s(x) = \nabla_x \log p(x),$$

that is, the gradient of the log-density. This vector field points toward regions of higher probability mass and therefore guides samples back toward the data manifold.

Directly estimating $s(x)$ is generally infeasible because $p(x)$ is unknown. Denoising Score Matching (DSM) [16] overcomes this difficulty by perturbing real data with Gaussian noise:

$$\tilde{x} = x + \sigma\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I),$$

and training a neural network to reconstruct the score of the perturbed distribution:

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}\left[\| s_\phi(\tilde{x}, \sigma) + \tfrac{\varepsilon}{\sigma} \|^2\right].$$

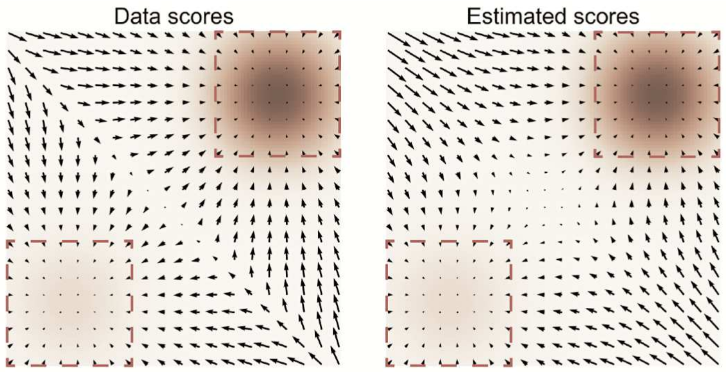

Figure 2.8: Denoising Score Matching (DSM). The model learns the score field $s_\phi(\tilde{x}, \sigma)$ of the noise-perturbed distribution by predicting the direction in which a corrupted input $\tilde{x}$ should move to return toward regions of higher probability. The arrows illustrate the score field, which always points toward areas of greater data density (adapted from Song Ermon, 2019).

Building on DSM, Noise Conditional Score Networks (NCSN) [15] extend score estimation to a hierarchy of noise levels. Sampling is then performed through annealed Langevin dynamics, a stochastic process alternating score-based updates with injected Gaussian noise:

$$x_{n+1} = x_n + \eta_t\, s_\phi(x_n, \sigma_t) + \sqrt{2\eta_t}\, \varepsilon_n.$$

As the noise level σ_t decreases, the updates become more deterministic and guide the samples toward the modes of the data density.

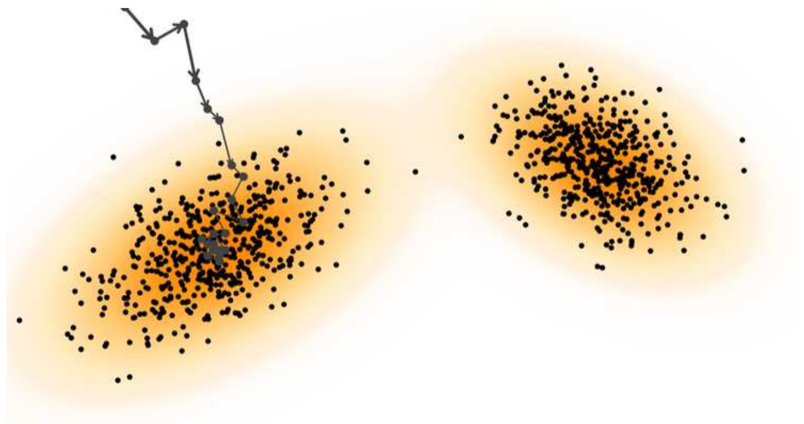

Figure 2.9: Annealed Langevin dynamics. Starting from an arbitrary initial state x_L, the sampler alternates injected Gaussian noise with gradient-ascent steps along the learned score field. As the noise level σ_t decreases, the dynamics become increasingly deterministic, guiding the trajectory toward regions of high data density.

A crucial conceptual result is that score-based models and diffusion models are mathematically equivalent [15]. In particular, predicting the noise $\varepsilon^*(x, \sigma)$ in DDPMs is equivalent to estimating the score of the noisy data:

$$s^*(x, \sigma) = -\frac{\varepsilon^*(x, \sigma)}{\sigma}.$$

Thus, denoising, noise prediction, and score estimation become alternative parameterizations of the same underlying log-density gradient.

This connection becomes even more explicit when diffusion models are formulated in continuous time. Recent extensions model the forward noising process as a Stochastic Differential Equation (SDE) [15]:

$$dx = f(x, t)\,dt + g(t)\,dW_t,$$

and show that the generative process corresponds to the solution of a reverse-time SDE. Every SDE is associated with a deterministic Ordinary Differential Equation, the Probability Flow ODE, whose trajectories share the same marginal distributions:

$$\frac{dx}{dt} = f_{ODE}(x, t).$$

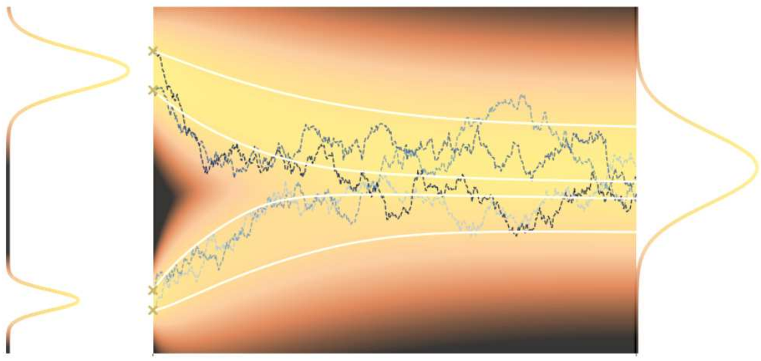

Figure 2.10: Forward diffusion expressed as a Stochastic Differential Equation (SDE). Trajectories starting from the data distribution p_{data} are gradually diffused by the drift and diffusion terms, and their marginals evolve toward a tractable prior distribution $p_T \approx p_{prior}$. This continuous-time view unifies diffusion models and score-based methods.

This unified perspective reveals that diffusion models, score-based models, and their SDE formulations are deeply interconnected. In particular, it shows that stochastic diffusion processes admit a fully deterministic counterpart, described by the Probability Flow ODE. This observation provides the conceptual foundation for generative models that operate without injected noise. Cold Diffusion, presented in the next section, emerges directly from this deterministic interpretation.

2.7 From Diffusion to Cold Diffusion

Classical diffusion models rely on stochastic Gaussian noise [8]. Cold Diffusion [2, 3] challenges this assumption, replacing randomness with *deterministic* degradations:

$$x_t = f_t(x_0),$$

where f_t may represent blurring, masking, downsampling, or desaturation. A neural network g_θ is trained to invert the degradation via

$$\mathcal{L} = \mathbb{E}_{x_0,t}\left[\|x_0 - g_\theta(f_t(x_0), t)\|^2\right].$$

Cold Diffusion preserves the iterative refinement philosophy of diffusion models while offering full interpretability of the forward process. Its deterministic nature aligns it with the Probability Flow ODE, making it a natural formulation for image restoration tasks where corruption is structured rather than stochastic [15].

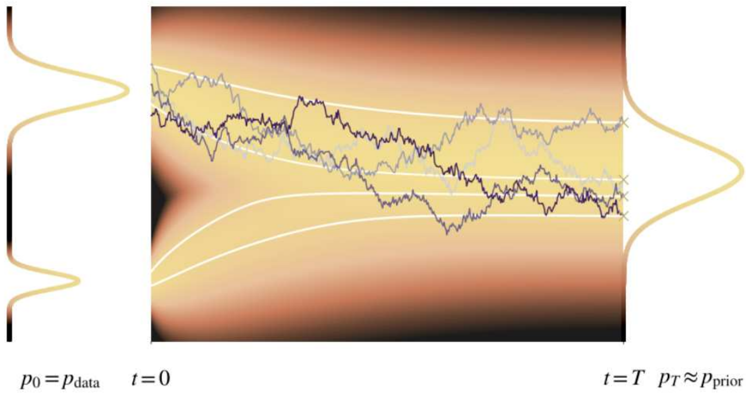

Figure 2.11: Reverse diffusion described through the reverse-time SDE, together with its associated deterministic Probability Flow ODE. Both processes share the same marginal distributions, but the Probability Flow ODE removes stochasticity, yielding deterministic trajectories. Cold Diffusion corresponds precisely to this deterministic limit, where generation is achieved without injected noise.

Although classical Variational Diffusion Models adopt a noise-centric formulation based on stochastic perturbations, Cold Diffusion instead embraces an intrinsically *operator-centric* viewpoint. In this setting, the forward trajectory is determined not by injected noise but by a structured, deterministic degradation operator f_t, and the neural network is trained to invert this transformation directly.

This perspective clarifies the conceptual role of Cold Diffusion within the unified theory of diffusion and score-based models. By eliminating stochasticity, it corresponds to the deterministic limit of the Probability Flow ODE, where generation arises through the inversion of a known degradation process rather than through the reversal of a stochastic noising mechanism. In this sense, Cold Diffusion preserves the incremental refinement characteristic of diffusion models while providing a transparent and fully interpretable formulation suited to restoration tasks.

Chapter 3
The Cold Diffusion Framework

This chapter presents the theoretical foundations of the Cold Diffusion framework, a deterministic alternative to classical diffusion models. While standard diffusion procedures rely on stochastic Gaussian perturbations to construct a Markov chain that progressively destroys the data, Cold Diffusion replaces randomness with fully deterministic degradation operators. This modification preserves the iterative refinement philosophy of diffusion models while eliminating stochasticity, yielding a forward process that is both precisely characterizable and fully interpretable. In the following sections, I formalize the structure of the Cold Diffusion framework, introduce the learning objective that enables the inversion of complex degradations, and discuss the stability improvements offered by the residual correction mechanism. Finally, I compare Cold Diffusion with related diffusion-based generative models, highlighting both conceptual similarities and fundamental differences.

3.1 Introduction

Traditional diffusion models, introduced in Chapter 2, are built on the idea that a generative process can be obtained by reversing a noisy degradation trajectory. This framework is supported by two main theoretical perspectives:

1. stochastic dynamics, specifically Langevin Dynamics;
2. probabilistic modeling through Variational Inference.

Both viewpoints justify the explicit use of Gaussian noise as a core element of the forward process, either as the driver of stochastic exploration or as a mathematically convenient choice to obtain tractable Markov transitions.

Cold Diffusion [2] challenges this assumption. The authors demonstrate that diffusion-like generative behavior can emerge even when the forward process is entirely deterministic. In their framework, noise is replaced with deterministic degradations, while the iterative structure of the model remains intact.

In this chapter, I present the general formulation of the Cold Diffusion framework, discuss its connection to classical diffusion theory, and explain how a generative model can operate without any stochastic component.

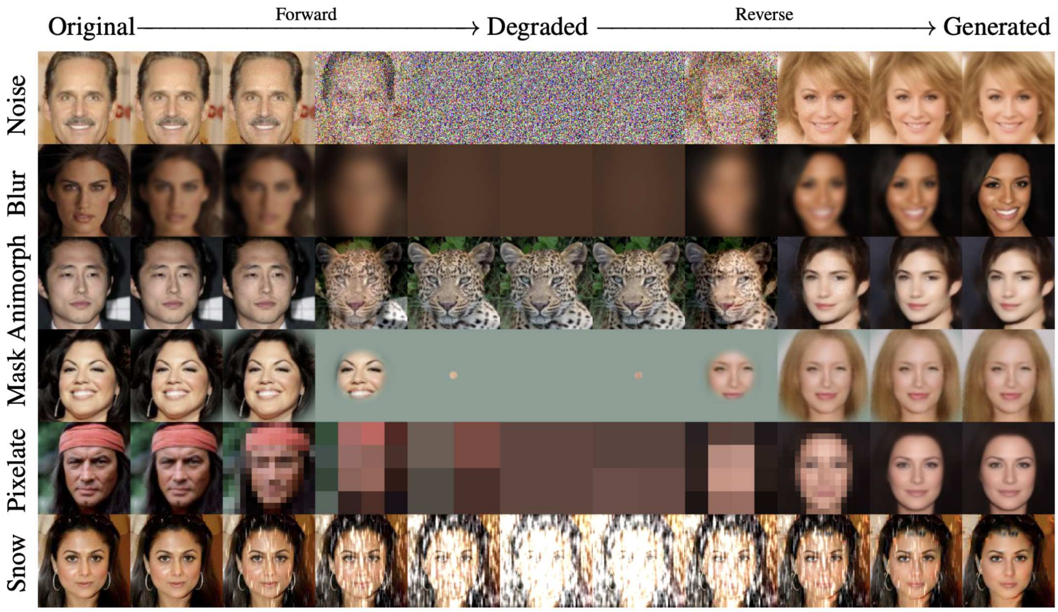

Figure 3.1: Examples of forward and reverse processes in Cold Diffusion. The forward process applies deterministic degradations such as blur, masking, pixelation, or snowification, while the reverse process reconstructs coherent images through the learned inversion operator. Adapted from Bansal et al. (2022).

3.2 Why Diffusion Traditionally Requires Noise

3.2.1 Langevin Dynamics and Score-Based Generation

The first justification for the use of noise comes from statistical physics. Langevin Dynamics describes the evolution of a variable according to:

$$x_{t+\Delta t} = x_t + \eta \, \nabla_x \log p(x_t) + \sqrt{2\eta} \, \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, I), \tag{3.1}$$

a formulation rooted in classical stochastic processes and closely related to score matching [9].

In this formulation:

- the term $\nabla_x \log p(x_t)$ pushes samples toward regions of higher probability density;
- the Gaussian noise prevents the dynamics from collapsing and ensures sufficient exploration of the space.

Score-based generative models, such as NCSN, are directly grounded in this idea [15]. They learn the score function $\nabla_x \log p(x)$ and use it within a noisy sampling procedure based on Langevin updates [16]. In this view, noise is an essential component: without it, the sampling trajectory would not behave as intended, and the model would become unstable.

3.2.2 Variational Inference and Gaussian Markov Chains

A second justification for noise arises from variational inference. DDPMs define the forward process as a Gaussian Markov chain [8]:

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t; \; \sqrt{1 - \beta_t} \, x_{t-1}, \; \beta_t I\right), \tag{3.2}$$

where $\{\beta_t\}$ is a variance schedule controlling the noise added at each step.

This design choice is crucial because:

- the Markov chain is analytically tractable;
- the marginal distribution admits a closed form:

$$q(x_t \mid x_0) = \mathcal{N}\big(x_t;\ \sqrt{\bar{\alpha}_t}\, x_0,\ (1 - \bar{\alpha}_t)I\big), \qquad \bar{\alpha}_t = \prod_{s=1}^{t}(1 - \beta_s); \tag{3.3}$$

- the training objective follows directly from a computable ELBO [11, 8].

Within this probabilistic interpretation, noise is not optional: it is what makes the chain mathematically manageable and enables the definition of a tractable reverse distribution $p_\theta(x_{t-1} \mid x_t)$.

Even though the reverse process is learned, the analytic structure of the forward process depends entirely on the Gaussian perturbations. Removing noise breaks the variational derivation, this is exactly the point questioned by the Cold Diffusion framework [2, 3].

3.2.3 Continuous Formulation: SDEs and the Probability Flow ODE

In the continuous-time limit, diffusion models can be described using stochastic differential equations. When the number of diffusion steps T tends to infinity, the discrete forward process of DDPMs converges to a stochastic differential equation (SDE) [15, 8]:

$$dx = f(x, t)\, dt + g(t)\, dW_t, \tag{3.4}$$

where W_t is a Wiener process. This formulation captures the continuous injection of Gaussian noise that progressively destroys the structure of the data.

Each such SDE has a corresponding deterministic equation, known as the Probability Flow ODE [15]:

$$dx = f_{PF}(x, t)\, dt. \tag{3.5}$$

The Probability Flow ODE shares the same marginal distributions as the SDE at every time t, but it removes the stochastic component entirely. As a result, sampling from this ODE yields a deterministic reconstruction trajectory, even though the forward process remains noisy. Since ODE trajectories are deterministic and therefore invertible, the Probability Flow ODE provides a mathematically tractable deterministic analogue of stochastic diffusion.

This connection motivates the conceptual shift behind Cold Diffusion: if the reverse dynamics can be deterministic without losing generative power, then the entire diffusion framework might be reformulated without relying on noise.

3.3 Deterministic Formulation of Cold Diffusion

Cold Diffusion replaces the noisy forward process with a deterministic degradation operator. Instead of adding Gaussian noise, the model applies a transformation that progressively corrupts the input [2, 3]:

$$x_t = D(x_0, t), \tag{3.6}$$

where $D(\cdot, t)$ is any deterministic operator satisfying:

- progressive degradation as t increases,
- continuity with respect to t,

- the identity condition $D(x, 0) = x$,
- no requirement of analytical invertibility.

Bansal et al. experiment with several types of deterministic degradations [2, 3], including:

- Gaussian blurring with increasing kernel variance;
- deterministic inpainting using Gaussian masks;
- recursive downsampling (super-resolution degradation);
- progressive colour desaturation;
- snowification (ImageNet-C distortions);
- deterministic morphing between animal and human faces (AFHQ).

In this framework, the forward process becomes a fully deterministic sequence of degradations, with no stochastic noise involved. A neural network learns to invert each degradation step, effectively reconstructing x_{t-1} from x_t.

Cold Diffusion therefore presents a deterministic reinterpretation of the diffusion paradigm, showing that iterative degradation and reconstruction, not stochasticity, are the essential components underlying diffusion-based generation [2].

3.4 The Reverse Problem: Learning to Invert D

In Cold Diffusion, the central task is to recover the original clean image from a deterministically degraded version [2, 3]. Since the forward process is simply the application of the operator $D(x_0, t)$, the reverse process must learn to undo this degradation. A neural network R_θ is therefore trained to predict the clean sample directly:

$$R_\theta(x_t, t) \approx x_0. \tag{3.7}$$

The loss function is a simple reconstruction objective [2]:

$$\mathcal{L} = \mathbb{E}_{x,t} \left\| R_\theta(D(x, t), t) - x \right\|_1. \tag{3.8}$$

Unlike stochastic diffusion models:

- the model does *not* predict noise,
- it does *not* optimize an ELBO or likelihood bound,
- and it relies on no explicit probabilistic formulation.

The task is purely to invert the deterministic transformation.

3.5 Inverse Sampling: Algorithm 1 and Algorithm 2

3.5.1 Algorithm 1 - Naive Reversal

A straightforward way to sample is to iteratively apply the learned inverse. Given a degraded state x_s, the model predicts an estimate of the clean image:

$$\hat{x}_0 = R_\theta(x_s, s), \tag{3.9}$$

and applies a degradation of lower severity:

$$x_{s-1} = D(\hat{x}_0, s-1). \tag{3.10}$$

This method assumes that R_θ acts as a near-perfect inverse of D, which is generally unrealistic. Small reconstruction errors accumulate over time, causing a drift from the ideal trajectory and producing visible artifacts. For this reason, Algorithm 1 is simple but unstable [2].

3.5.2 Algorithm 2 - Residual Correction Operator

To improve stability, Bansal et al. introduce a residual correction update:

$$x_{s-1} = x_s - D(\hat{x}_0, s) + D(\hat{x}_0, s-1). \tag{3.11}$$

This formulation:

- removes the degradation corresponding to level s,
- reinstates the degradation at level $s-1$,
- compensates for inaccuracies in the estimate $\hat{x}_0$.

The resulting trajectory remains closer to the ideal deterministic path and is significantly more stable.

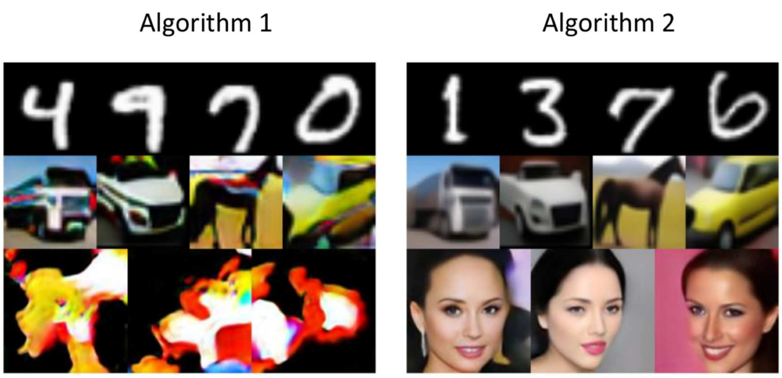

Figure 3.2: Comparison between Algorithm 1 (naive reversal) and Algorithm 2 (residual correction). Algorithm 1 accumulates errors rapidly and fails on complex datasets, whereas Algorithm 2 stabilizes the inversion by removing the degradation at step t and reapplying it at step $t-1$. Adapted from Bansal et al. (2022).

3.5.3 Motivation via Taylor Expansion

The authors justify Algorithm 2 using a first-order Taylor approximation [2]. For sufficiently smooth degradations:

$$D(x, s) \approx D(x, s-1) + \partial_s D(x, s-1). \tag{3.12}$$

Algorithm 2 cancels the first-order term, maintaining alignment with the degradation curve and reducing sensitivity to small reconstruction errors.

Intuitively, removing the first-order term stabilizes the reconstruction trajectory because it prevents the inversion dynamics from amplifying small local errors. In practice, the term $D(\hat{x}_0, t) - D(\hat{x}_0, t-1)$ acts as a local correction on the estimated clean image: it removes the portion of degradation that corresponds to the current step t and reinstates only the part associated with $t-1$. Since the mismatch between $\hat{x}_0$ and the true x_0 propagates through the derivatives of D, this difference compensates precisely for the first-order error accumulated in the estimate. As a result, Algorithm 2 remains aligned with the ideal degradation trajectory and prevents error amplification across successive steps.

In Algorithm 1, these errors accumulate at each step, since the model directly attempts to undo the degradation without correcting for local curvature. By subtracting the first-order contribution in Algorithm 2, the update becomes less sensitive to local fluctuations of the degradation operator, producing a smoother and more stable trajectory that remains closer to the underlying data manifold.

3.5.4 Connection to Deterministic ODEs

The deterministic forward trajectory induced by the degradation operator $D(x, t)$ can be viewed as the solution of an ordinary differential equation

$$\frac{dx(t)}{dt} = F(x(t), t),$$

whose flow defines a smooth degradation path from clean data to a heavily corrupted state.

Interpreting the process in this way clarifies the role of Algorithm 2. When sampling, one effectively integrates this ODE *backwards* in time. Algorithm 1 attempts this inversion directly, analogously to applying an explicit Euler step, and is therefore highly sensitive to local approximation errors.

Algorithm 2, by contrast, introduces a residual correction term that cancels the first-order deviation of the degradation operator. This makes the update behave like a more stable numerical integration scheme, reducing error accumulation and keeping the reconstructed trajectory closer to the true ODE flow.

3.6 Comparison Between Cold Diffusion and DDPM/DDIM

To better understand the contribution of Cold Diffusion, it is useful to compare its structure with classical diffusion models such as DDPM and DDIM [8, 15]. Although Cold Diffusion removes the stochastic component, many architectural and conceptual similarities remain [2, 3].

Table 3.1: Comparison between traditional diffusion models (DDPM/DDIM) and Cold Diffusion.

Aspect	DDPM / DDIM	Cold Diffusion
Forward process	Gaussian noise injection	Deterministic degradation
Formalism	ELBO, variational inference, Markov chain	Operator inversion
Training objective	Noise prediction	Image reconstruction
Sampling	Stochastic (DDPM) or deterministic (DDIM)	Fully deterministic
Theory	SDE / Probability Flow ODE	Implicit deterministic ODE
Stability	Ensured by injected noise	Residual correction stabilizes trajectory (Algorithm 2)
Complexity	Many iterative steps	Same number of steps, conceptually simpler

A further point of comparison is DDIM [14], which introduces a deterministic sampling procedure that parallels the Probability Flow ODE of score-based models. Although DDIM removes stochasticity during sampling, its forward process still relies on Gaussian noise and a predefined variance schedule. In contrast, Cold Diffusion is deterministic in both the forward and backward directions: no noise is used at any stage, and the degradation operator can be arbitrary rather than tied to a diffusion variance schedule. This difference underscores that Cold Diffusion is not a reformulation of DDIM, but a distinct framework in which the generative mechanism is driven entirely by the structure of the degradation operator rather than by stochastic diffusion.

Overall, Cold Diffusion demonstrates that noise is not inherently required for a diffusion-like generative mechanism. What truly matters is the presence of an iterative structure and the ability to learn an approximate inversion of a degradation process. This perspective broadens the landscape of diffusion methods and suggests that useful generative behaviors can arise from a wider class of deterministic transformations.

Chapter 4
Implementation of the Cold Diffusion Framework

This chapter presents the complete practical implementation of the Cold Diffusion framework introduced in Chapter 2. The goal is to translate the deterministic forward operators and their associated learned inverse mappings into two fully operational restoration pipelines: (i) a colour-desaturation process and (ii) a Gaussian deblurring process. These operators were chosen because they instantiate two conceptually distinct forms of degradation: chromatic suppression and geometric smoothing, thus providing complementary test cases for evaluating deterministic diffusion.

The implementations follow the formulation originally proposed by Bansal et al. [2], but are adapted for efficient training and evaluation within the constraints of Google Colab. To ensure methodological consistency, the chapter is organised into the following components: dataset preparation, definition of the deterministic operators, model architecture, training strategy, and iterative reconstruction. Each section details the practical considerations and design choices required to obtain a stable and fully functional Cold Diffusion pipeline.

4.1 Dataset Preparation

The dataset preparation follows a minimal yet rigorous preprocessing pipeline tailored to the requirements of the deterministic operators introduced in this chapter.

All experiments were conducted using the CelebA-HQ dataset, which provides high-quality and spatially aligned facial imagery. Following the protocol outlined in Chapter 3, each original 256×256 image was resized to 128×128 and linearly mapped to the range $[-1, 1]$ using

$$x \mapsto 2\left(\frac{x}{255}\right) - 1.$$

The resolution 128×128 offers a favourable balance between spatial fidelity and computational feasibility, enabling efficient training within the hardware constraints of Google Colab.

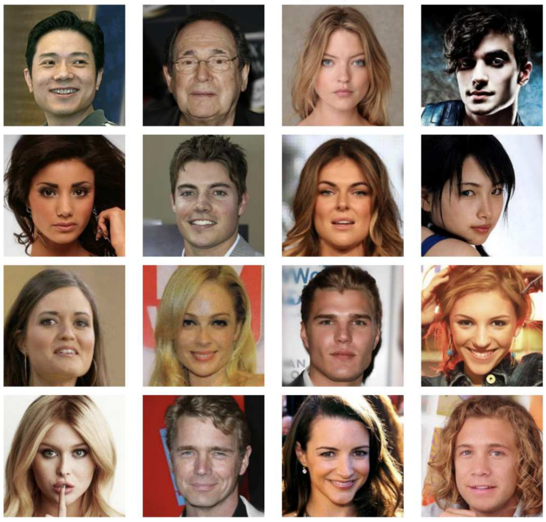

Figure 4.1: Representative samples from the CelebA-HQ dataset used in the experiments. The figure provides a visual overview of the facial images that constitute the training data.

Images were loaded dynamically at training time through PyTorch `DataLoader` structures (batch size 8), without storing any precomputed degradation trajectories. This avoids unnecessary storage overhead and ensures that each degraded sample remains synchronized with its corresponding reconstruction target, maintaining consistency across both the colour-desaturation and Gaussian deblurring pipelines.

4.2 Deterministic Degradation Operators

A defining characteristic of Cold Diffusion is that the forward process is deterministic rather than stochastic, in contrast to classical diffusion models [12, 8]. For each clean image x_0, a trajectory

$$x_t = D(x_0, t), \qquad t = 0, \dots, T-1,$$

is generated by progressively increasing the corruption strength. Both operators used in this thesis satisfy the identity constraint $D(x_0, 0) = x_0$, are differentiable, and evolve monotonically with t.

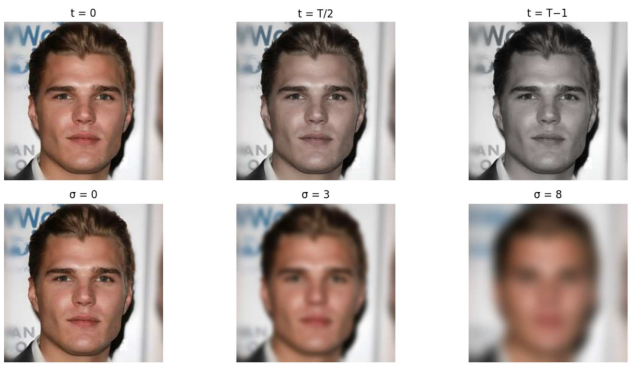

Figure 4.2: Deterministic forward operators used in this thesis. Top: Colour-desaturation operator at time steps $t = 0$, $t = T/2$, and $t = T-1$. Bottom: Gaussian blur operator with increasing blur intensity, parametrised by the standard deviation σ of the Gaussian kernel (here $\sigma \in \{0, 3, 8\}$).

4.2.1 Colour-Desaturation Forward Process

The first operator progressively suppresses chromatic information. Given an intermediate state x_{t-1}, the next state x_t is obtained as

$$x_t = (1 - \alpha(t))\, x_{t-1} + \alpha(t)\, \mathrm{gray}(x_{t-1}),$$

where $\mathrm{gray}(\cdot)$ denotes the per-pixel luminance and

$$\alpha(t) = \alpha_{\max} \frac{t}{T-1}, \qquad \alpha_{\max} = 0.7.$$

For $t = 0$ the operator reduces to the identity, whereas for $t = T - 1$ the image approaches a nearly grayscale configuration. This degradation primarily removes colour rather than geometric detail, making it substantially more invertible than blur-based operators. As a result, it provides a useful baseline for studying how Cold Diffusion behaves when the forward process remains structurally close to the original image while still exhibiting a clear, monotonic corruption trajectory.

4.2.2 Gaussian Blur Forward Process

The second operator implements a progressively increasing Gaussian blur. For each time index t, the blur level is parameterised by

$$\sigma(t) = \sigma_{\min} + (\sigma_{\max} - \sigma_{\min}) \frac{t}{T-1}, \qquad \sigma_{\min} = 0.4,\ \sigma_{\max} = 2.0.$$

In this thesis we fix $T = 6$, following the implementation described in the original Cold Diffusion work [2]. This choice also reflects a practical trade-off: for larger values of T, consecutive blur levels become increasingly similar and provide limited additional trajectory resolution, while smaller values of T reduce the granularity of the degradation schedule and make the inversion problem less stable. The setting $T = 6$ therefore offers a balanced progression of blur intensities that remains computationally efficient while preserving meaningful differences between successive degradation steps.

The corresponding 7×7 kernel K_t is generated *per sample* and *per channel*, and applied through grouped convolution:

$$x_t = K_t * x_0.$$

This produces smooth degradation trajectories with a strictly increasing amount of blur, while keeping the forward pass computationally lightweight. Higher values of $\sigma(t)$ increasingly suppress high-frequency details, making this operator a substantially more challenging testbed for deterministic reconstruction.

Figure 4.3: Examples of the Gaussian blur forward operator at different time steps t. Here the blur level follows the schedule $\sigma(t)$ defined in the text, producing a progressively stronger smoothing effect as t increases.

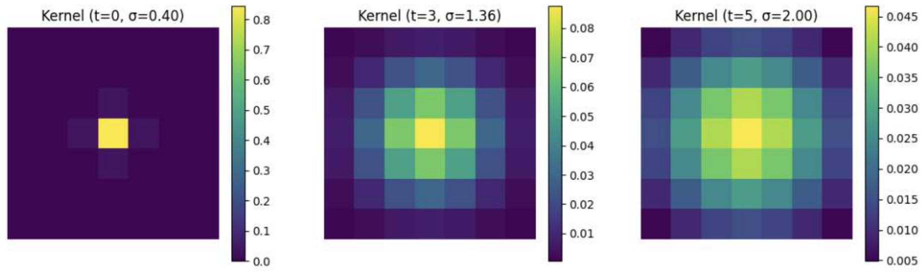

Figure 4.4: Gaussian kernels used in the forward degradation process for three different blur levels. As t increases, the kernel becomes increasingly diffuse, smoothing a larger spatial region.

4.3 Neural Architecture

Both operators share a U-Net-based reconstruction model $R_\theta(x_t, t)$, consistent with the architectures commonly employed in diffusion-based restoration [8]. In all experiments, the model receives as input the concatenation of the corrupted image x_t and a time-encoding map

$$\gamma(t) = \frac{t}{T-1},$$

broadcast to match the spatial resolution of the image.

Both U-Net variants were implemented directly in PyTorch and remain lightweight enough to be trainable on a Google Colab T4 GPU.

4.3.1 U-Net for Colour Desaturation

For the desaturation operator we employ a lightweight U-Net (approximately 0.6 million parameters), structured as follows:

- an initial convolutional block operating at full spatial resolution,
- two downsampling stages, each implemented via max-pooling followed by a convolution–ReLU block,
- a bottleneck block with increased channel dimensionality,
- two upsampling stages implemented with bilinear interpolation or transposed convolutions,
- skip connections linking encoder and decoder features at matching spatial resolutions.

This compact configuration was chosen to balance expressiveness and computational efficiency, given the simpler nature of the colour-collapse degradation.

4.3.2 UNetDeblur for Gaussian Blur

The deblurring operator requires recovering high-frequency content that is severely attenuated by the forward process. For this reason, we adopt a slightly deeper architecture, denoted `UNetDeblur`, featuring:

- an initial convolutional block taking 4 channels (RGB + time map),
- two encoder stages with max-pooling,

- a deeper bottleneck block,
- two decoder stages with transposed convolutions and skip connections,
- a final 1×1 convolution generating a three-channel output.

This model architecture aligns closely with the one used in the official Cold Diffusion implementation [2], while remaining lightweight enough to train on Colab GPUs.

4.4 Training Procedure

The training setup follows directly from the theoretical formulation introduced in Chapter 3. For each clean sample x_0, a time index t is drawn uniformly, and the corrupted image $x_t = D(x_0, t)$ is computed on the fly. The reconstruction model is then trained to predict the original image through the objective

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0,t} \, \|R_\theta(x_t, t) - x_0\|_1.$$

We use the Adam optimiser with a learning rate of 2×10^{-4} and a batch size of 8. Mixed-precision (AMP) training is employed whenever a GPU is available, improving throughput without affecting convergence.

For the colour-desaturation operator, the model is trained for 4000 iterations. For the deblurring operator, training is performed over four epochs on an 8000-image subset of CelebA-HQ. In this latter case, the sampling range for t is restricted to $t \in \{1, \dots, T-1\}$, since the case $t = 0$ corresponds to the identity transformation and provides no useful supervision for learning the inverse mapping.

4.5 Iterative Reconstruction

Once the reconstruction model is trained, Cold Diffusion provides two inversion strategies for recovering x_0 from a corrupted state x_t.

Naive inversion: At each step, the model predicts $\hat{x}_0 = R_\theta(x_t, t)$ and the forward operator is reapplied:

$$x_{t-1} = D(\hat{x}_0, t-1).$$

While conceptually simple, this approach tends to accumulate error along the trajectory, especially when the degradation is strongly lossy.

Residual-correction inversion: Following Bansal et al. [2], a more stable update rule is obtained by compensating for the discrepancy between the estimated $\hat{x}_0$ and the true signal:

$$x_{t-1} = x_t - D(\hat{x}_0, t) + D(\hat{x}_0, t-1).$$

This rule is essential for the Gaussian deblurring operator, where the information loss between successive blur levels can be substantial. In practice, reconstruction proceeds from $t = T-1$ down to $t = 1$, with $t = 0$ representing the final estimate of the clean image.

The combination of deterministic forward processes, U-Net-based inverse mapping, and residual-correction reconstruction produces two complete and fully functional instances of the Cold Diffusion framework. A detailed quantitative and qualitative assessment of their performance is reported in Chapter 5, including comparisons against classical deblurring baselines (Wiener filtering and sharpening), and metrics such as PSNR, SSIM, LPIPS [18], and FID.

These implementations constitute fully deterministic reconstruction pipelines, analysed thoroughly in the following chapter.

4.6 Classical Deblurring Baselines

In order to contextualise the performance of Cold Diffusion, two classical non-learned deblurring methods are included as baselines. Both are applied to images degraded with the same 15×15 Gaussian kernel ($\sigma = 2.0$) used in the forward process of the Cold Diffusion deblurring operator.

Wiener-like Filter: A frequency-domain Wiener deconvolution implemented using the known Gaussian kernel. This method attempts to invert the blur by suppressing high-frequency attenuation, but it is highly sensitive to noise amplification and typically fails to recover fine texture.

Sharpening (OpenCV): A standard spatial-domain sharpening filter implemented via `cv2.filter2D` using the kernel

$$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{pmatrix}.$$

The filter enhances edges but cannot reconstruct information lost through Gaussian smoothing. It serves as a simple non-learned point of comparison.

These baselines provide a reference point for evaluating the benefits of learning an operator-specific inverse mapping, as proposed in the Cold Diffusion framework.

4.7 Computational Considerations

The architectures employed in this thesis are deliberately compact to ensure efficient training under the hardware constraints of Google Colab. Each U-Net contains approximately 0.5–5 million parameters, depending on the operator configuration. Training with batch size 8 and diffusion depth $T = 6$ requires only a few hours per model.

Inference is similarly lightweight: reconstructing a single image requires only six forward passes through the network, leading to inference latencies on the order of tens of milliseconds on a modern GPU.

In contrast, classical DDPMs typically require several hundred reverse-diffusion steps. Deterministic diffusion therefore reduces the computational burden by one to two orders of magnitude, trading full generative sampling capability for a restoration-oriented design aligned with deterministic operators.

Chapter 5
Results and Discussion

This chapter presents the quantitative and qualitative results obtained from the two deterministic diffusion pipelines implemented in this work: (i) the colour-desaturation (DeColor) operator, and (ii) the Gaussian deblurring operator. The evaluation follows the methodology established in Chapter 4, and the interpretation of the results is grounded in the theoretical framework developed in Chapter 3.

We assess reconstruction quality using standard metrics in image restoration: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), Learned Perceptual Image Patch Similarity (LPIPS) [18], and Fréchet Inception Distance (FID). In all experiments, the reconstruction follows the iterative inversion rule proposed by Bansal et al. [2], using the residual-correction update described in Section 4.5.

5.1 Evaluation Metrics

PSNR: The Peak Signal-to-Noise Ratio measures the fidelity of reconstruction in terms of pixel-wise error. Higher values indicate a lower mean-squared reconstruction error.

SSIM: The Structural Similarity Index captures luminance, contrast, and structural consistency between the reconstructed and original images, correlating strongly with perceptual quality.

LPIPS: LPIPS [18] uses deep feature distances to quantify perceptual similarity. Lower values correspond to more perceptually accurate reconstructions.

FID: The Fréchet Inception Distance evaluates the distributional similarity between sets of real and reconstructed images. Lower values indicate that reconstructions lie closer to the manifold of natural images.

All metrics are evaluated on held-out subsets of CelebA-HQ, using the protocols specified for each restoration task.

5.1.1 Alternative Evaluation: Classifier-Based Assessment

Although pixel-wise and perceptual metrics (PSNR, SSIM, LPIPS, FID) provide a comprehensive evaluation of reconstruction quality, an additional perspective can be obtained through classifier-based assessment. This methodology, originally introduced in the context of GAN evaluation, trains a classifier exclusively on reconstructed images and tests it on real data, thereby measuring whether the reconstructed samples preserve class-discriminative information. Approaches of this type have been explored in works such as InfoGAN[5] and BigGAN[4] , where the ability to retain or reproduce semantic structure plays a central role.

In the context of Cold Diffusion, classifier-based evaluation can serve as a complementary diagnostic tool. Since deterministic inversion aims to recover not only global appearance but also the underlying semantic consistency of the input image, testing classification accuracy across reconstructed samples would allow us to quantify whether the inversion process preserves identity, facial attributes, or other high-level features. This is particularly relevant in settings where pixel-based metrics may not fully capture semantic fidelity, such as in highly blurred or partially irreversible degradations.

Although not employed in this thesis, classifier-based assessment represents a promising direction for future analysis. Integrating such a metric would enable a more complete characterisation of the extent to which deterministic diffusion preserves both low-level structure and high-level semantic information, complementing the pixel-level and distributional metrics presented in the following sections.

5.2 Results: Colour-Desaturation (DeColor)

The DeColor operator progressively suppresses chromatic information while preserving the spatial structure of the image. Given the mild and largely invertible nature of this degradation, the Cold Diffusion framework performs exceptionally well.

The evaluation was conducted on 80 randomly sampled images using the residual-correction reconstruction (Algorithm 2). Table 5.1 reports the quantitative metrics.

Table 5.1: Reconstruction metrics for the DeColor Cold Diffusion operator (80 images, Algorithm 2).

Method	PSNR ↑	SSIM ↑	LPIPS ↓
Cold Diffusion (DeColor)	**38.01**	**0.986**	n/a

Note. LPIPS is not reported as a single global value because the colour-desaturation operator is nearly invertible, yielding LPIPS scores close to zero across the dataset. Instead, the per-step behaviour of LPIPS along the deterministic degradation schedule is illustrated in Figure 5.1, which provides a more informative characterisation of perceptual stability.

While Table 5.1 summarises the overall reconstruction performance, Figure 5.1 examines how the metrics vary across the degradation schedule.

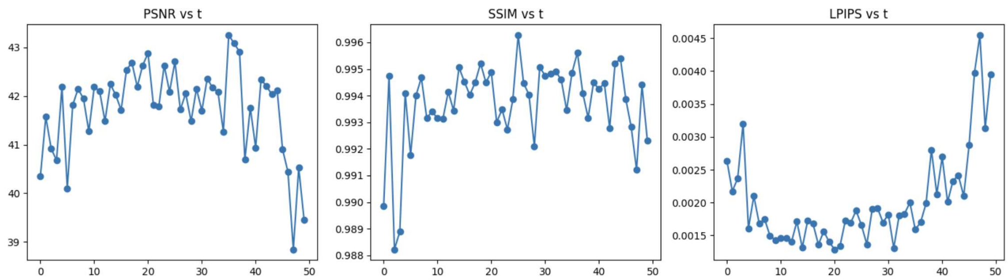

Figure 5.1: Mean PSNR, SSIM and LPIPS values computed across 50 test images for each degradation level t in the DeColor process. The metrics demonstrate the strong stability of the reconstruction across the deterministic trajectory, reflecting the near-invertibility of the colour-desaturation operator.

To further analyse the stability of the reconstruction along the deterministic trajectory, we compute PSNR, SSIM and LPIPS for each degradation level t. Figure 5.1 shows the mean behaviour of these metrics over 50 test images. As illustrated, reconstruction quality remains highly stable across the entire degradation chain, confirming that the DeColor operator induces very limited information loss and is therefore easily invertible by the model.

The results show a near-perfect reconstruction of chromatic content. The SSIM value of 0.986 indicates that structural information is almost fully preserved along the deterministic forward trajectory. PSNR above 38 dB is consistent with the low-information loss of the operator.

Figure 5.2: Qualitative example of deterministic colour-desaturation within the Cold Diffusion framework. From left to right: original clean image, degraded image obtained through the DeColor forward operator, and reconstructed output using Algorithm 2. The reconstruction accurately restores chromatic information, confirming the near-invertibility of the colour-suppression trajectory.

Qualitatively, the model reconstructs both low- and high-frequency colour details, with negligible colour drift. This behaviour is fully consistent with the observations made by Bansal et al. [2], who highlight that deterministic colour collapse is one of the most stable settings for Cold Diffusion.

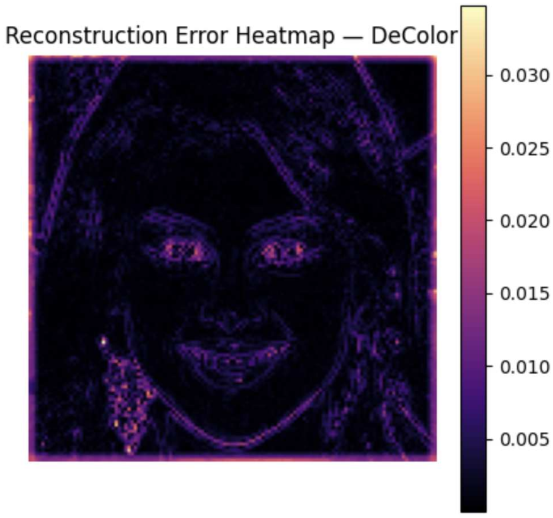

Figure 5.3: Pixel-wise reconstruction error heatmap $|x_0 - \hat{x}_0|$ for the DeColor Cold Diffusion operator. The error remains extremely low across most of the image, with slightly higher values concentrated along high-frequency edges (hair, eyes). This confirms that the colour-desaturation process induces minimal information loss and is therefore easily invertible by the model.

5.3 Results: Gaussian Deblurring

The Gaussian blur operator represents a substantially more challenging setting, as high-frequency texture is progressively suppressed and must be reconstructed from the learned inverse mapping. Following the experimental protocol described in Chapter 4, we evaluate reconstruction on a set of 200 test images using the iterative inversion Algorithm 2.

Table 5.2 summarises the global quantitative results.

Table 5.2: Reconstruction metrics for Gaussian deblurring via Cold Diffusion (200 images, Algorithm 2).

Method	PSNR ↑	SSIM ↑	LPIPS ↓
Cold Diffusion (Deblurring)	**28.67**	**0.894**	**0.1725**

Note. In this case, LPIPS is reported as a global metric because the blur operator induces perceptually meaningful distortions that remain non-negligible even after reconstruction, making an aggregate value informative for assessing perceptual fidelity.

To analyse performance along the reconstruction trajectory, Figure 5.4 reports the mean PSNR and SSIM values as a function of the degradation level t.

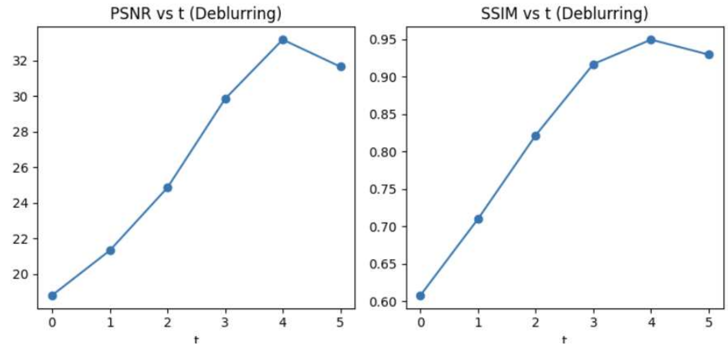

Figure 5.4: Mean PSNR and SSIM values across the reconstruction trajectory for the Gaussian deblurring operator. Both metrics increase steadily from $t = 0$ to $t = 4$, indicating progressive improvement in reconstruction quality. A slight decrease at $t = 5$ suggests that the final step introduces minor smoothing or over-correction effects, which marginally reduce both pixel-level and structural fidelity.

A complementary perspective is provided by the per-image dispersion of metrics. Figure 5.5 shows the relationship between PSNR and SSIM for the 200 reconstructed samples. The two metrics exhibit a clear positive trend: samples achieving higher pixel-wise fidelity also tend to display better structural similarity. The limited variability and absence of significant outliers suggest that the reconstruction model behaves consistently across the dataset.

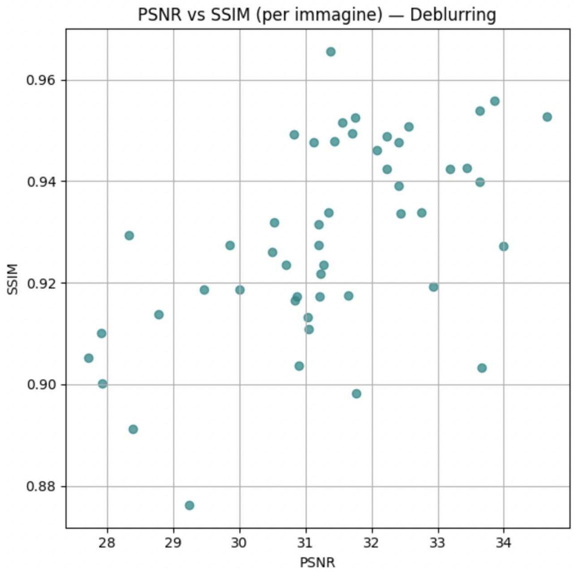

Figure 5.5: Scatter plot of PSNR versus SSIM computed across the 200-image test set for the Gaussian deblurring task. Higher PSNR values consistently correspond to better SSIM scores, indicating that pixel-level fidelity and structural coherence improve jointly. The limited dispersion suggests stable behaviour across the dataset.

Overall, the quantitative results indicate that the model successfully reconstructs global and mid-frequency structures, achieving SSIM values close to 0.9. An LPIPS score of 0.17 confirms perceptually coherent restoration, although subtle high-frequency texture remains challenging, especially under strong blur ($\sigma_{\max} = 2.0$).

To evaluate distributional alignment, we compute the Fréchet Inception Distance (FID) between reconstructed and real images:

$$\text{FID} = 64.02.$$

This value is consistent with results reported in [2], indicating a moderate distributional shift attributable to the inherent information loss introduced by the blur operator.

A qualitative example is shown in Figure 5.6. The model recovers global facial geometry and mid-frequency details while maintaining consistency with the statistics of the dataset.

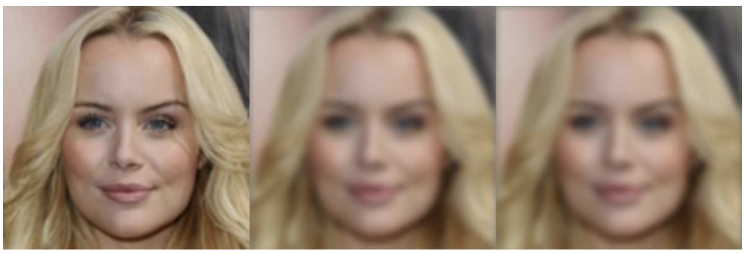

Figure 5.6: Qualitative example of Gaussian deblurring reconstruction using Cold Diffusion. From left to right: original clean image, Cold Diffusion reconstruction (Algorithm 2), and the degraded input at the strongest blur level. The model recovers global structure and mid-frequency details while maintaining consistency with the dataset statistics.

To better characterise reconstruction behaviour across the dataset, we analyse the global distribution of pixel-wise absolute error. As shown in Figure 5.7, the distribution is highly skewed: the majority of pixels exhibit negligible reconstruction error, while a long-tailed component corresponds to regions where high-frequency information, such as hair, edges, and fine texture, is particularly difficult to recover. This behaviour is consistent with the limitations imposed by the Gaussian blur operator.

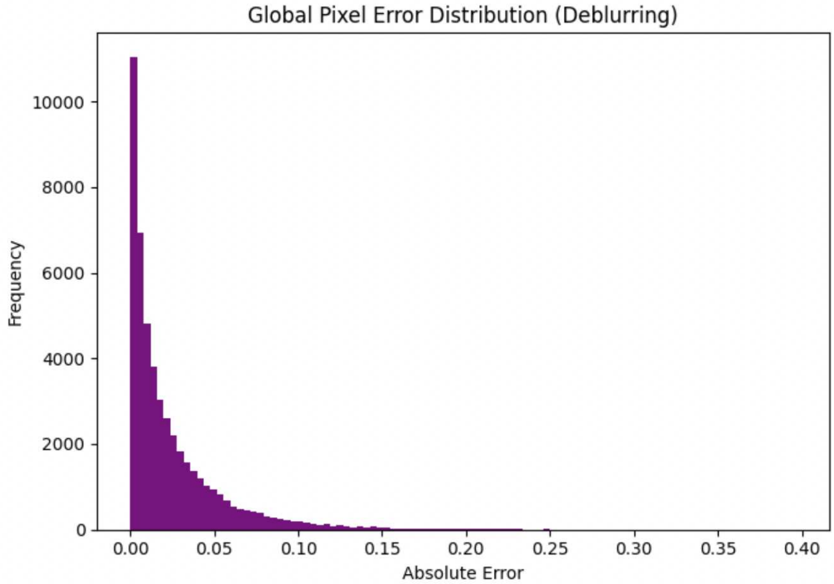

Figure 5.7: Global pixel-wise absolute error distribution for the Gaussian deblurring task. Most pixels exhibit near-zero reconstruction error, while a long tail corresponds to high-frequency regions such as hair, edges, and fine structural details. This reflects the inherent information loss induced by the blur operator.

5.4 Comparison with Classical Baselines

To contextualise the performance of Cold Diffusion, we compare it with two classical deblurring baselines: (i) a Wiener-like filtering approach, and (ii) a sharpening-based restoration method. All baselines are applied to images blurred with a 15×15 Gaussian kernel ($\sigma = 2.0$).

Table 5.3: Cold Diffusion vs. classical deblurring baselines (200 images).

Method	PSNR ↑	SSIM ↑	LPIPS ↓
Cold Diffusion (Deblurring)	**28.67**	**0.894**	**0.1725**
Wiener-like Filter	23.18	0.666	0.3868
OpenCV Sharpen	28.36	0.850	0.1931

Cold Diffusion decisively outperforms Wiener filtering across all metrics. Compared to the sharpen baseline, it achieves:

- higher SSIM (+0.044),
- lower LPIPS (better perceptual fidelity),
- slightly higher PSNR.

These results confirm the advantage of learning an operator-specific inverse mapping, as emphasised in [2]. Sharpening enhances edges but fails to reconstruct fine texture coherently, whereas Cold Diffusion leverages the training distribution to infer plausible high-frequency structure.

5.5 Qualitative Analysis

A qualitative inspection of the reconstructed samples reveals clear and consistent patterns across both deterministic operators. In the case of the DeColor process, the model restores chromatic information with remarkable precision, even at the highest levels of desaturation. Skin tones, hair chromaticity, and global colour appearance are recovered almost perfectly, with only negligible artefacts. This behaviour reflects the inherently mild and nearly invertible nature of the colour-suppression operator, which preserves spatial structure and removes only a limited amount of information from the input images. The qualitative findings are fully aligned with the quantitative metrics reported in Section 5.2, and with previous observations in the literature concerning the stability of colour-desaturation transforms. For the Gaussian deblurring operator, the reconstruction task is substantially more challenging, yet the model recovers global facial geometry and mid-frequency details with notable reliability. The overall shape, coarse features, and structural coherence of the faces are consistently restored. Nevertheless, some residual softness remains in regions with strong high-frequency content such as hair strands, eyebrows, and fine wrinkles-areas where the forward blur removes information that is intrinsically difficult to reconstruct, even for learned inversion models. This behaviour is perfectly consistent with the distribution of reconstruction errors shown in Figure 5.7 and with the moderate LPIPS and FID values obtained in this setting. Across both operators, the iterative inversion trajectory behaves stably. The residual-correction mechanism plays a central role in ensuring this stability, as it compensates for the discrepancies between predicted and true signals and prevents the accumulation of reconstruction errors across steps. This effect is particularly important in the deblurring scenario, where the information loss increases substantially between consecutive blur levels. Overall, the qualitative results convincingly illustrate the capability of Cold Diffusion to invert deterministic degradation processes while maintaining coherent visual structure throughout the reconstruction trajectory.

To complement the quantitative analysis presented in the previous sections, we report additional qualitative examples for both degradation operators. Each example includes the original image, the degraded observation, the output of a classical baseline, and the reconstruction produced by the proposed Cold Diffusion model. These examples provide a direct visual comparison that highlights the strengths and limitations of each method across a wider range of samples.

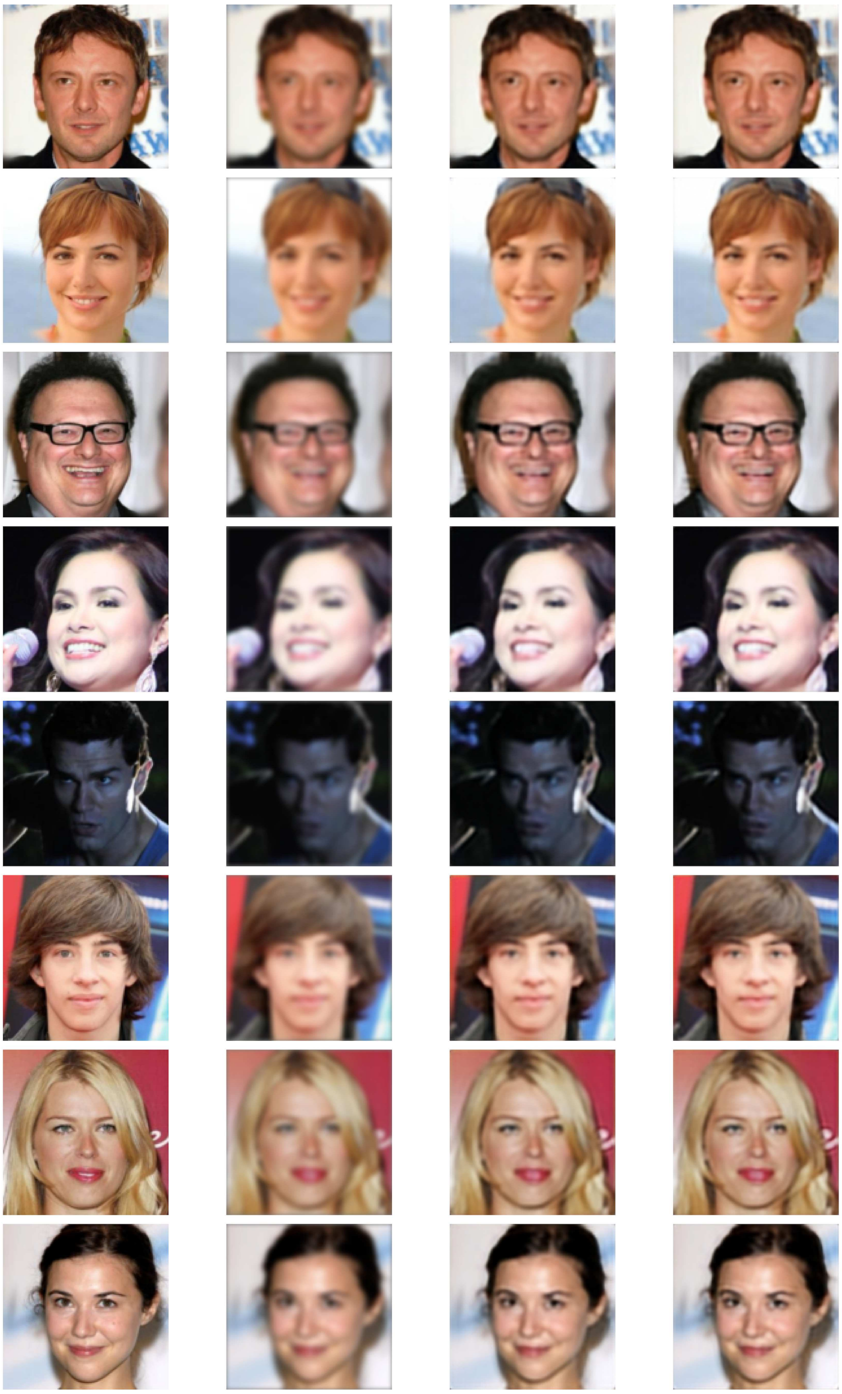

Figure 5.8: Qualitative comparison for the Gaussian deblurring task. Each row shows: (i) the original image, (ii) the degraded input, (iii) the Wiener baseline reconstruction, and (iv) the proposed Cold Diffusion output.

Figure 5.9: Qualitative comparison for the DeColor task across multiple samples. Each row shows: (i) the original clean image, (ii) the degraded observation produced by the colour-desaturation operator, (iii) the reconstruction obtained using a classical baseline method (OpenCV sharpening), and (iv) the output of the proposed Cold Diffusion model. The Cold Diffusion reconstruction restores chromatic information with high fidelity, closely matching the original colour distribution across all examples.

5.6 Discussion

The experiments demonstrate that Cold Diffusion is a flexible and effective framework for deterministic image restoration. Consistent with the theoretical insights of Bansal et al., the quality of reconstruction depends critically on the information loss induced by the forward operator.

- **DeColor: nearly invertible:** High PSNR and SSIM confirm that most information is retained along the forward trajectory, enabling highly accurate inversion.
- **Deblurring: partially invertible:** Although blur removes high-frequency information, the learned estimator successfully recovers plausible details, outperforming classical baselines.

The results validate the central claim of Cold Diffusion: explicit stochasticity is not necessary for high-quality iterative reconstruction.

To summarise the differences in reconstruction behaviour between the two deterministic operators, Figure 5.10 provides a direct quantitative comparison of PSNR and SSIM for the DeColor and Deblurring models. This cross-operator comparison highlights the critical role of forward-process invertibility in determining the achievable reconstruction quality.

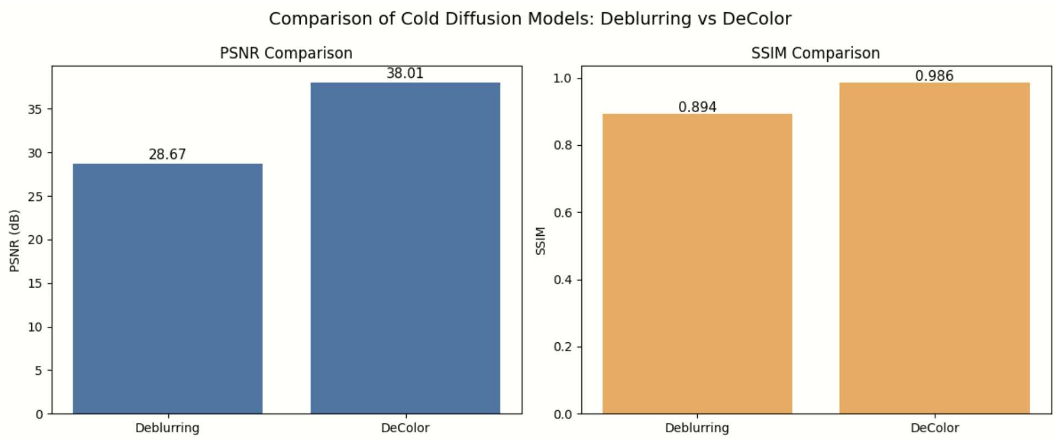

Figure 5.10: Quantitative comparison of Cold Diffusion models trained on the DeColor and Deblurring operators. The DeColor process, being nearly invertible, yields significantly higher PSNR and SSIM values, whereas the Gaussian blur operator introduces irreversible high-frequency loss that constrains reconstruction accuracy.

Deterministic operators, when paired with a learned inverse and a residual correction rule, can approximate the expressive power of stochastic diffusion models within restoration tasks.

5.7 Limitations and Threats to Validity

Despite the encouraging results, several limitations constrain the generalisability of the findings:

- **Restricted domain:** All experiments are conducted on aligned facial images from CelebA-HQ, a highly structured domain. Generalisation to heterogeneous or complex scenes remains unverified.

- **Simplified deterministic operators:** Only two degradations are explored: colour desaturation and isotropic Gaussian blur. More realistic degradations such as motion blur, JPEG artefacts, haze, or spatially varying kernels are not addressed.
- **Model and computational budget:** The compact U-Nets and limited dataset subset reflect practical constraints rather than an attempt to reach state-of-the-art performance.
- **Absence of semantic evaluation:** Evaluation focuses on PSNR, SSIM, LPIPS, and FID. Classifier-based or task-driven metrics, which assess semantic consistency, are not included and remain future work.

These limitations highlight the boundaries of the current study while suggesting clear opportunities for methodological refinement in future research.

Chapter 6
Conclusions and Future Work

This thesis explored both the theoretical foundations and the practical feasibility of Cold Diffusion, a deterministic reformulation of diffusion-based generative modelling in which classical Gaussian noise is replaced by structured, fully deterministic degradation operators. The work combined a conceptual overview of modern generative frameworks with the implementation of two complete Cold Diffusion pipelines: colour desaturation and Gaussian deblurring applied to the CelebA-HQ dataset. The overarching goal was to evaluate whether stochasticity is truly essential to the expressive power of diffusion models, or whether their characteristic behaviour can instead be reproduced through the inversion of noise-free transformation trajectories.

From a theoretical perspective, the thesis retraced the evolution of generative modelling from Autoencoders and Variational Autoencoders to Generative Adversarial Networks and stochastic diffusion models. This trajectory highlights a common theme: modern generative methods can be interpreted as forms of iterative refinement, where an initial perturbation of the data is progressively reversed. Within this view, the core generative capability of diffusion models does not arise from Gaussian randomness itself, but from the structured, multi-step evolution of data across a degradation trajectory. Cold Diffusion makes this principle explicit by removing noise while preserving the iterative structure typical of diffusion, naturally aligning with the deterministic interpretation of the Probability Flow ODE.

On the experimental side, the thesis implemented two deterministic operators exhibiting different levels of invertibility. The colour-desaturation operator, which collapses chromatic information while preserving spatial structure, proved almost entirely reversible: reconstructions achieved PSNR values above 38 dB and SSIM values close to 0.99, confirming that deterministic inversion is highly stable when the forward operator preserves most of the underlying information. The Gaussian blur operator, which progressively suppresses high-frequency detail and represents a substantially more challenging setting, required greater modelling capacity and benefited significantly from the residual-correction update. Even under this partially irreversible degradation, the model recovered coherent geometry and perceptually plausible texture, outperforming classical non-learned baselines such as Wiener filtering and sharpening. These observations indicate that deterministic diffusion remains effective across degradations of varying difficulty, provided that the reconstruction network is sufficiently expressive and the inversion dynamics are properly stabilised.

The conclusions drawn here are consistent with recent developments in the broader field of generative modelling. Following the introduction of Cold Diffusion, the community has progressively shifted toward deterministic or noise-free formulations of diffusion processes, including DDIM, the Probability Flow ODE, flow matching, and rectified flows. These approaches reinforce the idea that high-quality generative behaviour does not strictly require explicit noise injection. Related work has also explored learned forward operators, where the degradation trajectory is optimised jointly with the inversion network, further supporting the operator-centric interpretation adopted in this thesis. Studies such as Cold SegDiffusion and ColDBin confirm that deterministic degradation-inversion pipelines can extend well beyond image restoration, achieving strong performance in tasks like segmentation and document binarisation where consistency and interpretability are crucial.

Several promising directions for future work emerge from this study. Extending Cold Diffusion to more challenging or heterogeneous degradations such as motion blur, haze, JPEG artefacts, or irregular occlusions, would offer deeper insight into the limits of invertibility and the robustness of residual-correction dynamics. Hybrid architectures could combine deterministic operators with controlled stochasticity, leveraging the interpretability of deterministic models with the regularising benefits of noise. Another natural extension involves learning the forward operator jointly with the inverse mapping, allowing the model to discover degradation trajectories that maximise reconstruction quality. Moreover, including classifier-based or task-driven evaluation would enable the assessment of whether deterministic inversion preserves semantic or discriminative information beyond pixel-level metrics.

Another promising direction involves studying composite degradation operators, in which multiple transformations, such as colour-desaturation followed by Gaussian blur, are combined into a single deterministic trajectory. Real-world degradations rarely occur in isolation, and analysing the invertibility of such hybrid operators would provide a deeper understanding of how different sources of information loss interact. Exploring whether a single Cold Diffusion model can jointly invert heterogeneous degradations would offer valuable insight into the scalability and generality of deterministic diffusion frameworks.

Overall, this thesis contributes to clarifying both the conceptual foundations and the practical viability of deterministic diffusion models. The results demonstrate that deterministic degradation-inversion pipelines are not only feasible but capable of producing high-quality reconstructions across degradations of different severity. More broadly, the work positions Cold Diffusion as an early and influential formulation within a wider shift toward noise-free generative modelling. By challenging the long-standing assumption that stochasticity is essential to diffusion-based generation, it opens the way for new operator-driven frameworks that integrate restoration, generative modelling, and representation learning within a coherent and interpretable deterministic paradigm.

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